

# HS Data Science 26

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## Session 01: Orientation, Topics, and Workflow

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# Seminar Data Science

## Focus of This Year's Seminar

- retrieval-augmented generation
- agentic search
- open web retrieval on our own infrastructure

## Core Idea

This seminar rewards students who can formulate an interesting question, design a manageable study, and defend their choices clearly.

# Seminar Topic

The seminar focuses on retrieval-augmented generation and smart agents in the context of open web search.

## Technical Context

- we run our own search engine on the [Open Web Index](#)
- students can explore topics through this infrastructure
- the environment supports retrieval experiments and agent workflows

## Research Perspective

- search is part of the research object
- retrieval choices can be inspected and compared
- the seminar goes beyond a generic RAG demo

# Why OWI and Our Search Engine?

The OWI-based environment exists because web retrieval research is otherwise structurally constrained by proprietary search engines and black-box APIs.

## What This Enables

- inspect retrieval behaviour
- compare sparse, dense, and hybrid retrieval
- study provenance and ranking choices
- build agentic workflows on transparent search

## Consequence for the Seminar

- the search engine is not just a utility
- the retrieval setup itself can be analysed
- good topics exploit the openness of the system

## Demo Search

Use the [OURRS demo web search](#) for demonstrations and experiments. The API will be available there as well.

# How Topic Selection Works

You are expected to generate your own research direction within the seminar theme.

## Good Starting Angles

- retrieval quality
- agent planning and verification
- trust and usability
- corpus and query analysis

## What Makes a Good Topic

- interesting research question
- manageable scope
- clear evaluation strategy
- relevance to the seminar setup

## Example

- weak: *"Build a RAG demo"*
- better: *"When does query reformulation improve agentic search on exploratory tasks?"*

# Expected Seminar Work

## Scientific Working

- conduct paper research
- identify the gap or tension
- formulate research questions
- document assumptions and limits

## Methodology

- implement a prototype or workflow
- run small-scale experiments
- compare alternatives where useful
- explain why the evidence supports the claim

A modest experiment with a sharp question and clean reasoning is better than an oversized prototype without a defensible result.

# Grading

| Component                            | Weight |
|--------------------------------------|--------|
| Independent idea generation          | 20%    |
| Small-scale but rigorous experiments | 15%    |
| Clear methodology and evaluation     | 15%    |
| Presentation                         | 40%    |
| Report                               | 10%    |

The asymmetry is intentional: idea generation and presentation are the parts students cannot outsource blindly to AI.

# Deliverables

## Expected Outputs

- a 1-pager research narrative
- a seminar presentation with a clear research narrative
- a concise report in ACM format
- a Git repository with code, prompts, experiments, or analysis artefacts

## Practical Rule

Keep every artefact aligned with one central claim: what question you asked, how you investigated it, and what you learned.



# Organization

## Study Programmes

- BSc Computer Science
- BSc Internet Computing
- MSc Computer Science
- MSc AI Engineering
- open to other faculties

## Semester Rhythm

- orientation and topic exploration
- question development
- prototype implementation
- presentations with feedback
- refinement and final report

Lock the question early, then iterate on evidence and presentation quality.

# Tentative Schedule

| Date              | Topic                                   |
|-------------------|-----------------------------------------|
| Tue 14/04/26      | Topic Presentation, Research Hypothesis |
| Tue 21/04/26      | Scientific Working I                    |
| Tue 28/04/26      | Scientific Working II                   |
| Tue 05/05/26      | Scientific Working III                  |
| Tue 12/05/26      | Research Hypothesis Presentation        |
| Tue 19/05 – 23/06 | Q&A sessions                            |
| Tue 30/06 – 14/07 | Student Presentations                   |

All sessions: 10:00–12:00, ITZ R 138.



# Next Steps

Before the first week of semester, students should:

- research the field of RAG, agentic search, and web search
- write down one page with a possible topic or research direction
- submit it or present it in the first week of semester

Research-hypothesis presentations are 5–10 minutes.

# Looking Ahead

Before the next milestone, each student should prepare:

- three candidate topic ideas
- one tentative main research question
- one plausible evaluation strategy
- one short argument for why the question is interesting

This seminar is about producing an interesting, defensible piece of research in an AI-rich environment where originality, judgment, and communication matter more than routine implementation.